

PURNA MUMMANI

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Summary

Software Engineer with 3 years of experience designing, developing, and optimizing scalable, secure, and high-performance enterprise applications for global clients. Strong expertise in full-stack development, cloud-native architectures, API engineering, database optimization, CI/CD automation, and Agile delivery models. Proven ability in improving system reliability, reducing latency, and accelerating release cycles for platforms used by millions of users.

Professional Experience

Adobe

Jun 2025 – Present

Software Engineer Intern

Remote, USA

- Collaborated with senior engineers in the design and development of scalable **microservices** using **Java, Spring Boot, and React** for platforms supporting **2M+ active users**, contributing to backend optimizations that improved processing efficiency through refined **AWS (Lambda, EC2, RDS)** resource utilization.
- Strengthened security posture by developing **REST APIs** integrated with **OAuth 2.0 and JWT** for role-based access control; improved **PostgreSQL** query performance by **~28%** through strategic indexing and schema optimization.
- Supported DevOps workflows using **Jenkins, Docker, and GitHub Actions**, reducing manual deployment effort by **~30%** and cutting mean time to issue detection (MTTD) by **~25%** via **AWS CloudWatch** monitoring.
- Ensured production-grade code quality by conducting **peer code reviews** and implementing **secure development practices** (OWASP), maintaining high adherence to Adobe's internal engineering standards across the SDLC.

UNCC: Office of Assessment and Accreditation

May 2024 – Apr 2025

Graduate Research Assistant

Charlotte, NC

- Conducted data-driven research on student learning outcomes using **Python, Excel, and SQL**, performing statistical analysis and data visualization that contributed to a **~20% improvement** in institutional assessment strategies
- Developed interactive dashboards using **Tableau/Power BI** to automate grading analytics and reporting workflows, reducing instructor grade processing time by **~25%** and improving data-driven decision-making for academic leadership.

Accenture

Jan 2022 – Dec 2023

Software Engineer

Hyderabad, India

- Developed and maintained enterprise-grade backend applications using **Java, Spring MVC, and Hibernate** to support complex business workflows, processing **1M+ transactions per day** and reducing application downtime by **~28%** through modular fault isolation.
- Built full-stack interfaces using **Angular and RESTful services**, optimizing **MySQL and Oracle** database schemas to achieve **~34%** faster data retrieval and a **~22% reduction** in end-user support tickets.
- Automated CI/CD pipelines and testing using **JUnit and Mockito**, increasing test coverage to **85%** while migrating services to **AWS**, resulting in a **25% reduction** in infrastructure costs and a **30% improvement** in client satisfaction.

Technical Skills

Programming Languages: Java, Python, C++, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks: React, Node.js, Express.js, Angular, Spring Boot, MVC, JPA, Hibernate, PyTorch, OpenCV, FastAPI, Flask

Databases: MySQL, PostgreSQL, NoSQL (MongoDB), Redis

Developer Tools: Git, Github, Apache spark, Hadoop, Kafka, Maven, ZooKeeper, LangChain, Pinecone, Jupyter, TensorFlow, Google Colab, Tableau, JIRA, Confluence, Postman, BitBucket, Streamlit

Cloud Services: AWS(EC2, S3, RDS, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, GitHub Actions, Azure, CloudFormation, API Gateway

Core Competencies: Enterprise SaaS, Microservices, End-to-End System Design, CI/CD, Distributed Systems, Agile, , SDLC, RESTful APIs, Full Stack, End-to End development.

Certifications: AWS Certified Developer – Associate

Projects

Real-Time Video Action Detection System | *PyTorch, Python, Transformers, OpenCV*

- Developed a real-time action detection pipeline using **PyTorch, CNN-Transformer**, and **OpenCV** to identify and localize human actions in untrimmed videos with **~9.8% mAP improvement** over I3D baselines, and implemented a multi-resolution feature fusion to optimize computational efficiency by **85%**.
- Deployed the system on **AWS EC2** using **Python, FastAPI, AWS S3** and integrated **Lambda** for video preprocessing achieving **8 FPS** real-time detection and processing **2,000+** video clips daily under load-tested conditions.

University Carpooling Platform | *Full-stack*

- Designed a full-stack campus ride-sharing platform using **Java, Spring Boot, React**, hosted on **AWS EC2** enabling students to share rides with automatic cost-splitting and implemented **OAuth 2.0** authentication with student ID validation.
- Integrated **AWS API Gateway** with backend for secure API routing, designed **SQL** data models on **AWS RDS** for ride requests and **AWS S3** for profile images, ensuring efficient data management with automated notifications via **Twilio SMS**.

NextStep AI: AI-Powered Resume Matching Platform | *Agentic AI & Cloud-Native*

- Architected an AI job-matching agent using **Python & Ollama (Local LLMs)** with **SentenceTransformers** to generate semantic similarity analysis for resume-to-job alignment, reducing manual screening time by **~80%** and improving match accuracy through foundation model embeddings.
- Developed a dynamic backend pipeline in **Python** using **AWS SQS** and **Lambda**, integrated SerpAPI and **Indeed RSS** for real-time job retrieval on **AWS S3**, enabling asynchronous skill-gap analysis to optimize cost for targeted applications.

Education

University of North Carolina at Charlotte GPA: 3.8/4

Jan 2024 – Dec 2025

Master of Science in Computer Science

Charlotte, NC

Courses: Data Structures & Algorithms, Object Oriented Analysis, Database Systems, Cloud Computing, Software System Design and Implementation, Visual Analytics, Artificial Intelligence, Machine Learning and Distributed Systems